STATISTICALLY MODELING THE OLD SCHOOL RUNESCAPE HIGH SCORES

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DATA EXPLORATION AND EXTRACTION

- OSRS High scores
- A quantitative analysis of the Old School Runescape hiscores(2022).

LUKE AREND ON GITHUB

- Data were scraped for the top 2 million players on the OSRS hiscores. Data consists of xp, rank, and level in each OSRS skill and overall, along with rank and score stats for clue scrolls, minigames and bosses.
- Account data were deduplicated, sorted and subsampled to keep skill level columns only. After deduplication, 1999625 records remained. Each record is a length-23 vector giving an account's levels in the 23 OSRS skills.

.CSV

- player-stats-raw.csv
- 2007339 rows × 198 columns
- player-stats.csv
- 1999625 rows × 26 columns

SIZE REDUCTION

```python
# Ensuring reproducibility by setting a random seed
random.seed(42)

# Generate three random selections of 10,000 rows
for i in range(1, 4):
    sample_df = df.sample(n=10000, random_state=random.randint(0, 10000))  # Random seed changes for each file
    sample_df.to_csv(f'player_stats_sample_{i}.csv', index=False)
    print(f"Sample {i} saved as 'player_stats_sample_{i}.csv'")
```

```
Sample 1 saved as 'player_stats_sample_1.csv'
Sample 2 saved as 'player_stats_sample_2.csv'
Sample 3 saved as 'player_stats_sample_3.csv'
```

```python
# Load the sampled datasets
df1 = pd.read_csv('player_stats_sample_1.csv')
df2 = pd.read_csv('player_stats_sample_2.csv')
df3 = pd.read_csv('player_stats_sample_3.csv')
```

DATAFRAME 1

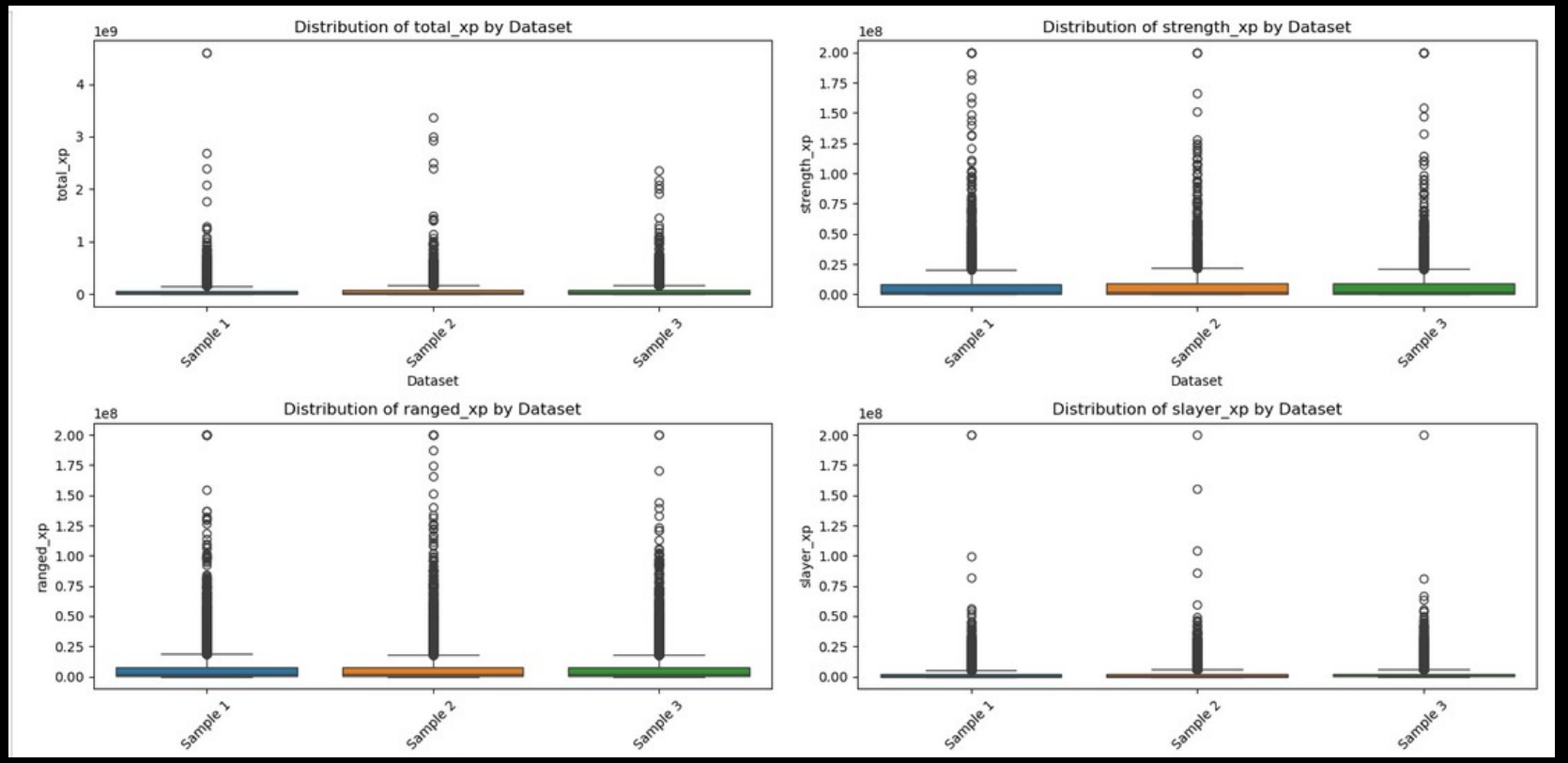
```
[10]: df1.describe(include = "all")
```

[10]:

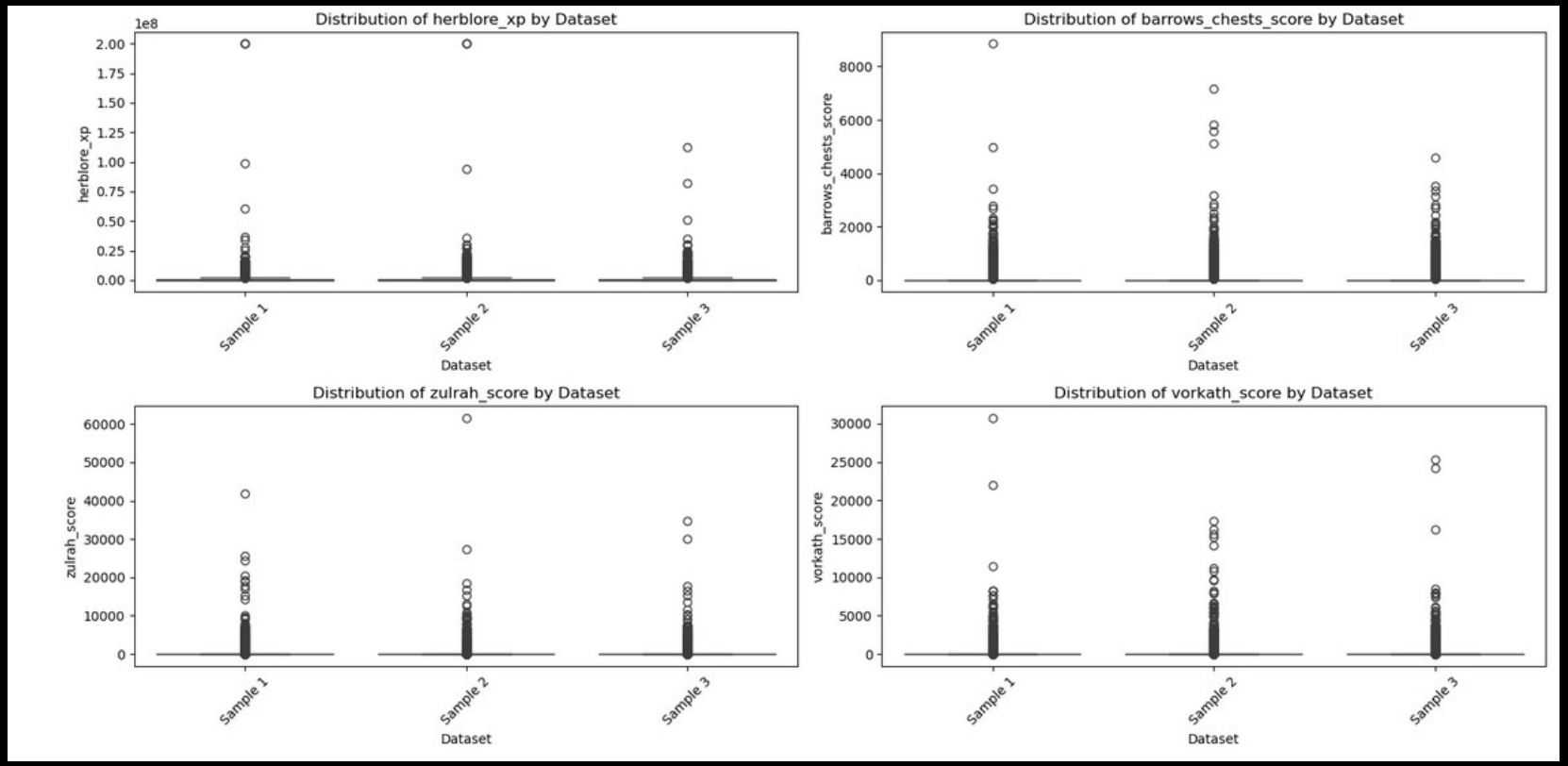
	username	total_rank	total_level	total_xp	attack_rank	attack_level	attack_xp	defence_rank	defence_level	defence_xp	...	vetion_scor
count	10000	1.000000e+04	10000.00000	1.000000e+04	1.000000e+04	10000.000000	9.997000e+03	1.000000e+04	10000.000000	9.982000e+03	...	10000.00000(
unique	9999	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN
top	Hiddy	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN
freq	2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN
mean	NaN	1.005892e+06	1424.67010	6.233747e+07	7.853244e+05	68.010900	4.071134e+06	7.297442e+05	63.969200	3.468804e+06	...	3.31590(
std	NaN	5.811232e+05	403.35253	1.237551e+08	6.040859e+05	29.440483	8.122788e+06	5.905013e+05	31.536755	6.977748e+06	...	96.49928
min	NaN	2.400000e+01	807.00000	8.689970e+05	-1.000000e+00	-1.000000	-1.000000e+00	-1.000000e+00	-1.000000	-1.000000e+00	...	-1.00000(
25%	NaN	5.030038e+05	1065.75000	8.746008e+06	2.143605e+05	61.000000	3.235830e+05	1.523065e+05	60.000000	2.738282e+05	...	-1.00000(
50%	NaN	1.012052e+06	1392.50000	2.465564e+07	7.430035e+05	75.000000	1.213460e+06	6.859375e+05	71.000000	8.695325e+05	...	-1.00000(
75%	NaN	1.509058e+06	1753.00000	7.036275e+07	1.281353e+06	87.000000	4.013912e+06	1.202393e+06	85.000000	3.352316e+06	...	-1.00000(
max	NaN	1.999952e+06	2277.00000	4.600000e+09	1.999897e+06	99.000000	2.000000e+08	1.999682e+06	99.000000	2.000000e+08	...	6462.00000(

11 rows × 198 columns

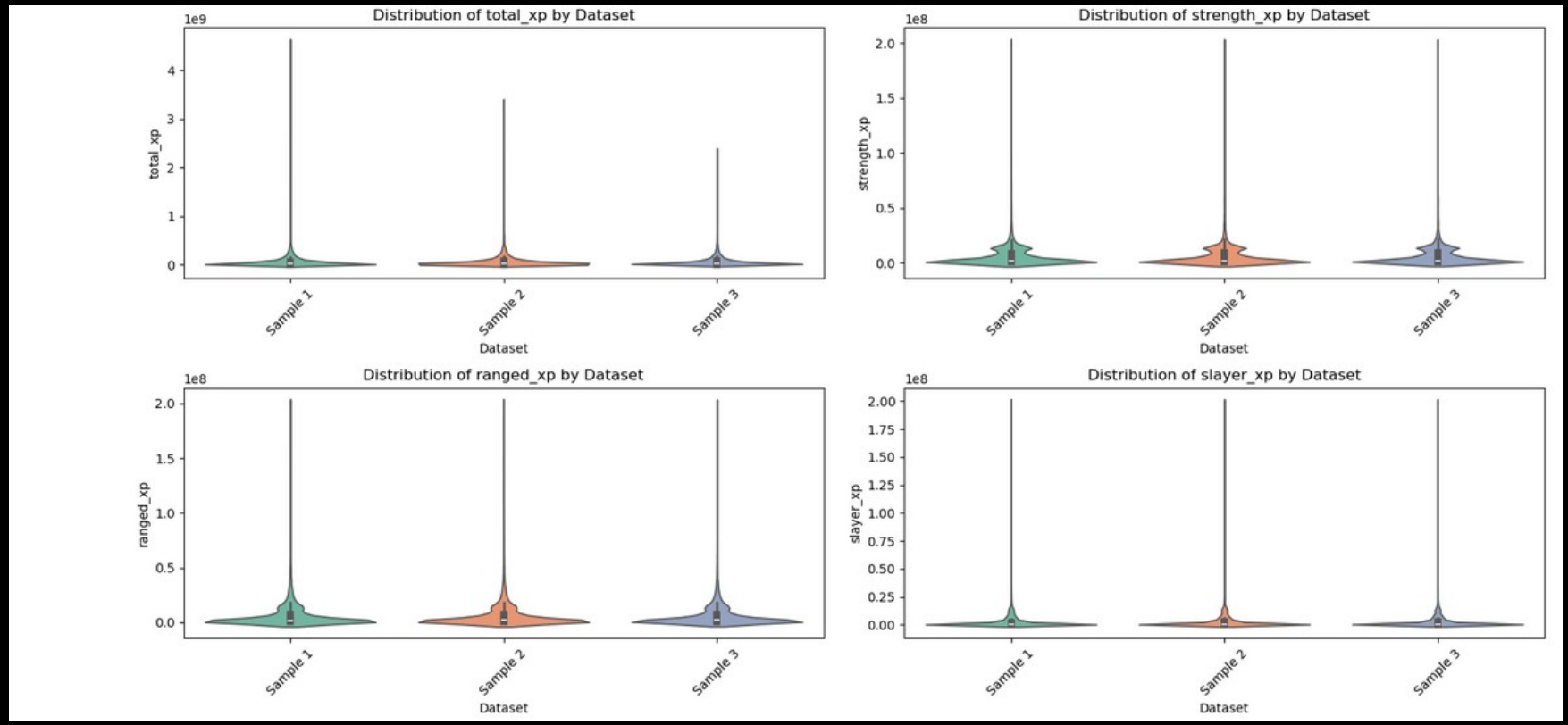
COMPARING SAMPLES



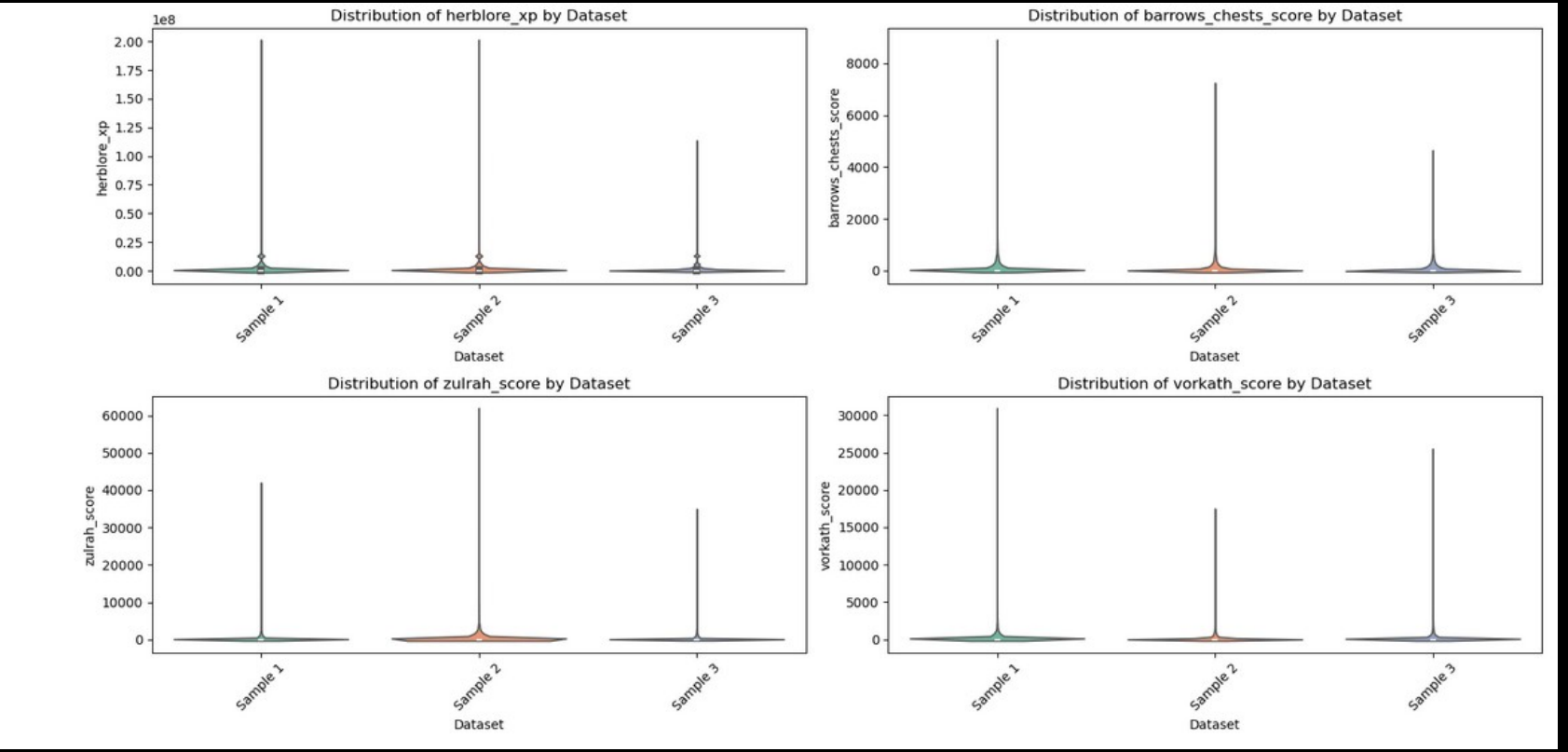
COMPARING SAMPLES CONT. 1



COMPARING SAMPLES CONT. 2



COMPARING SAMPLES CONT. 3



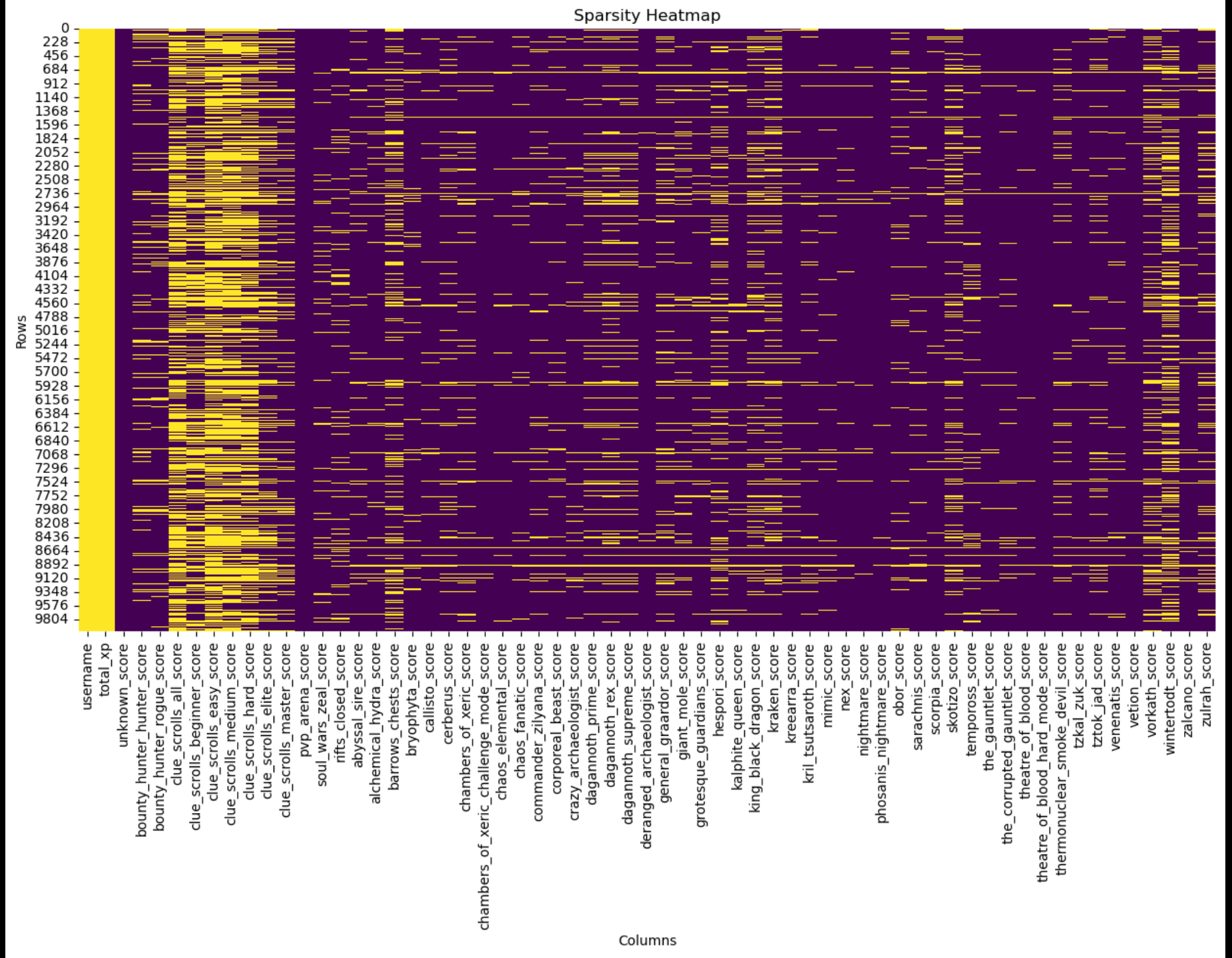
DATA CLEANING

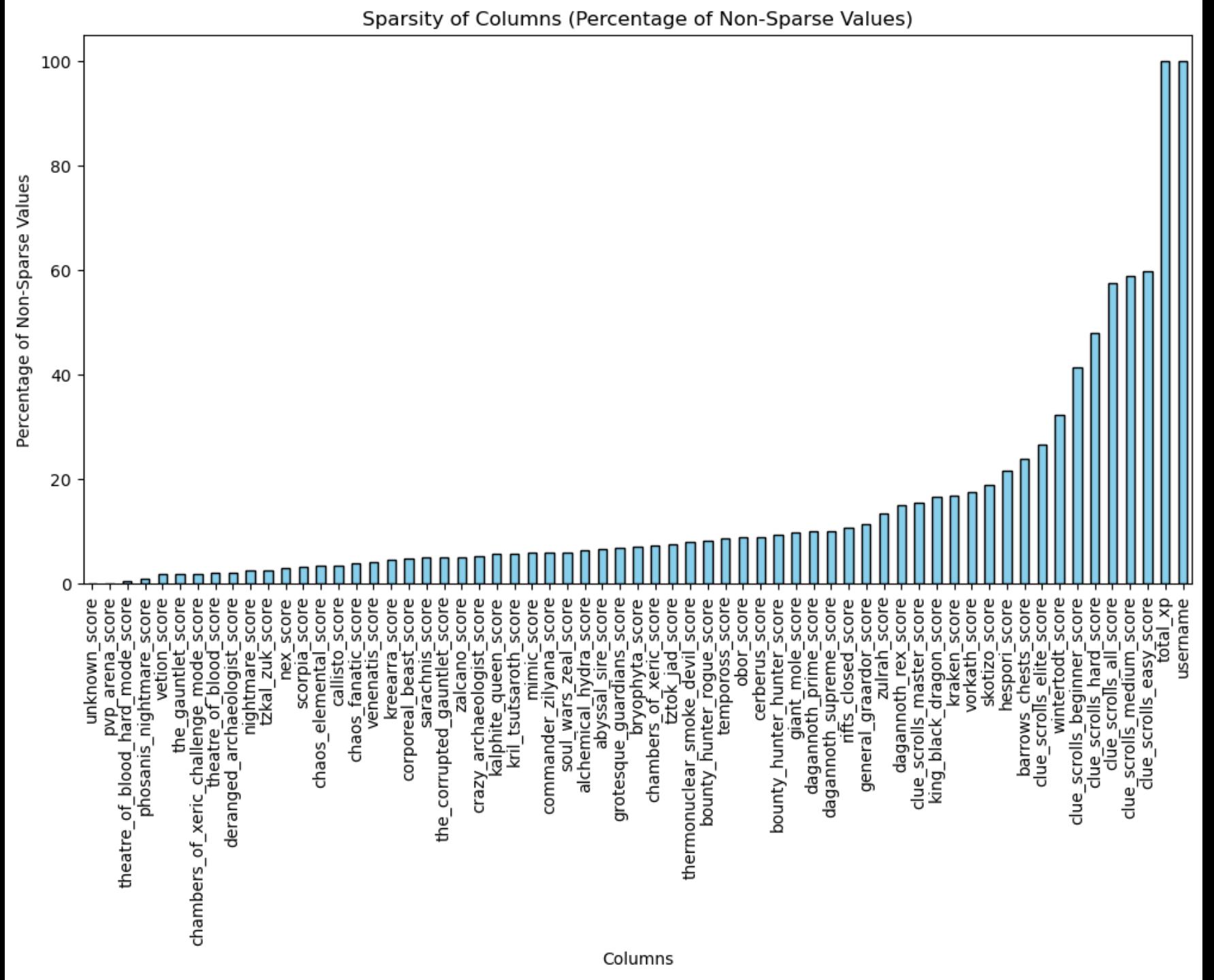
- Dropped all 'rank' columns
- Dropped 'timestamp'
- Dropped 'username'
- Dropped 'dataset'
- Dropped all 'level' and 'xp' columns
- Kept 'total_xp' and all columns with 'score' in them.

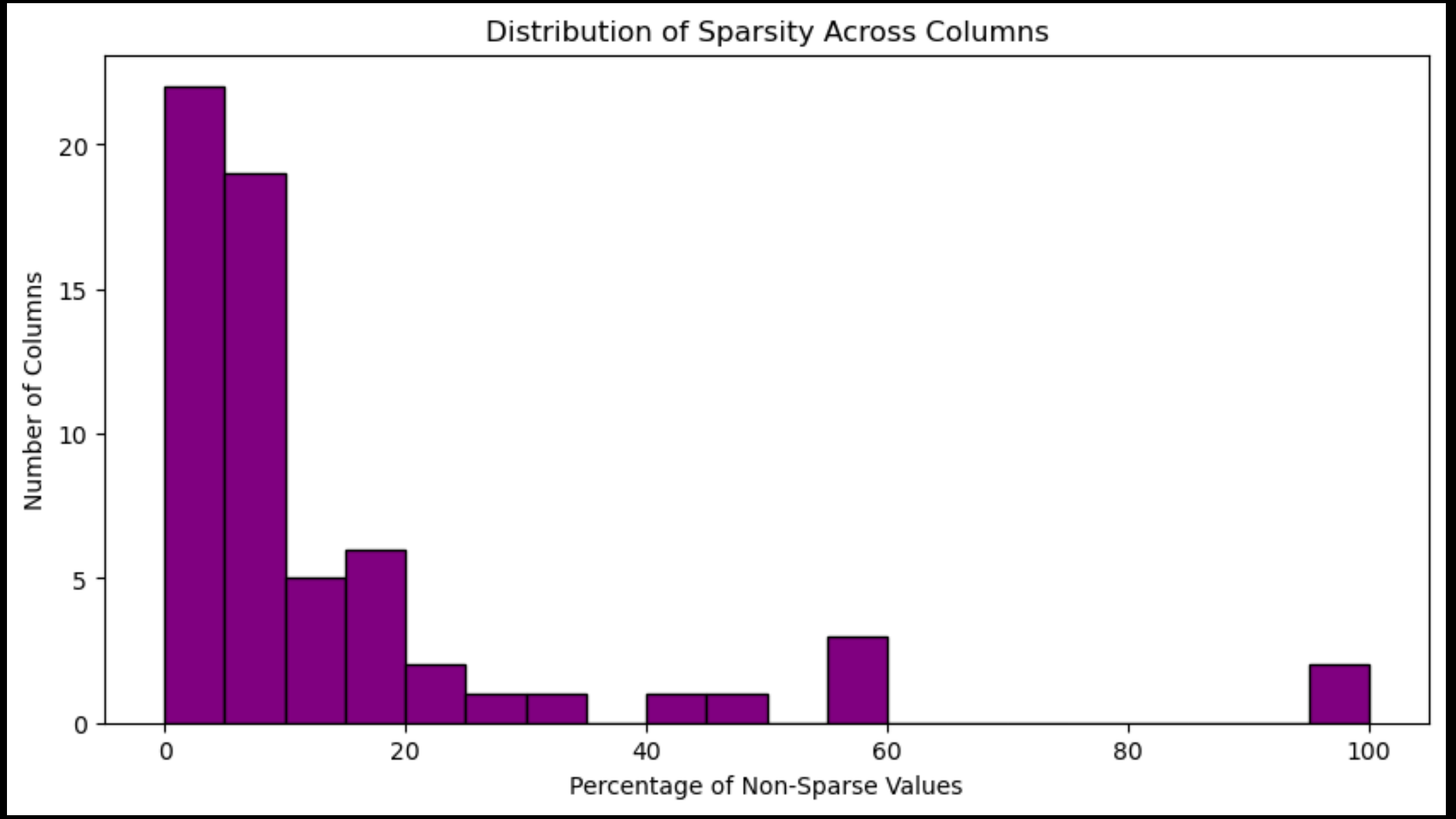
DF1.HEAD()

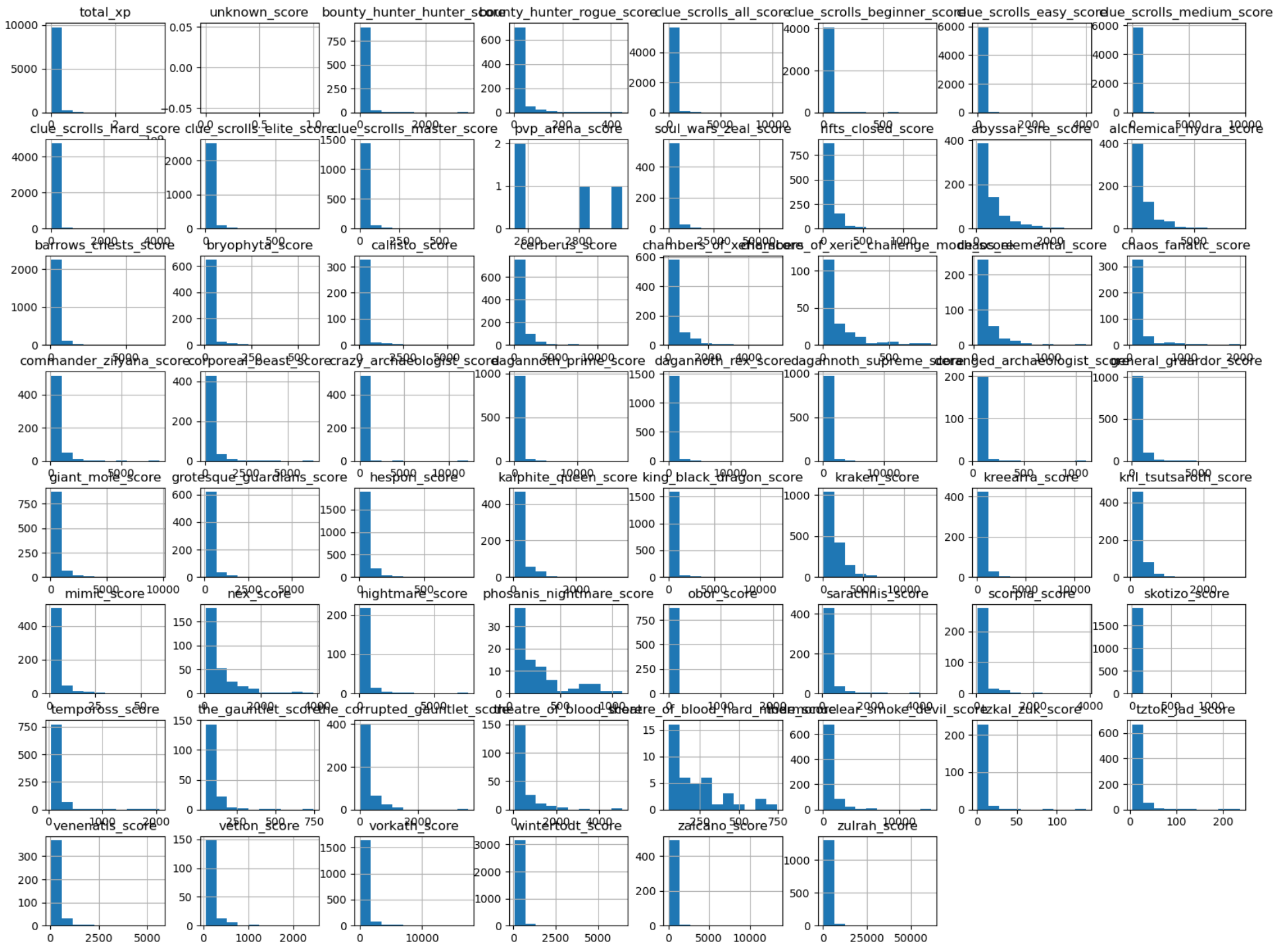
[70]:

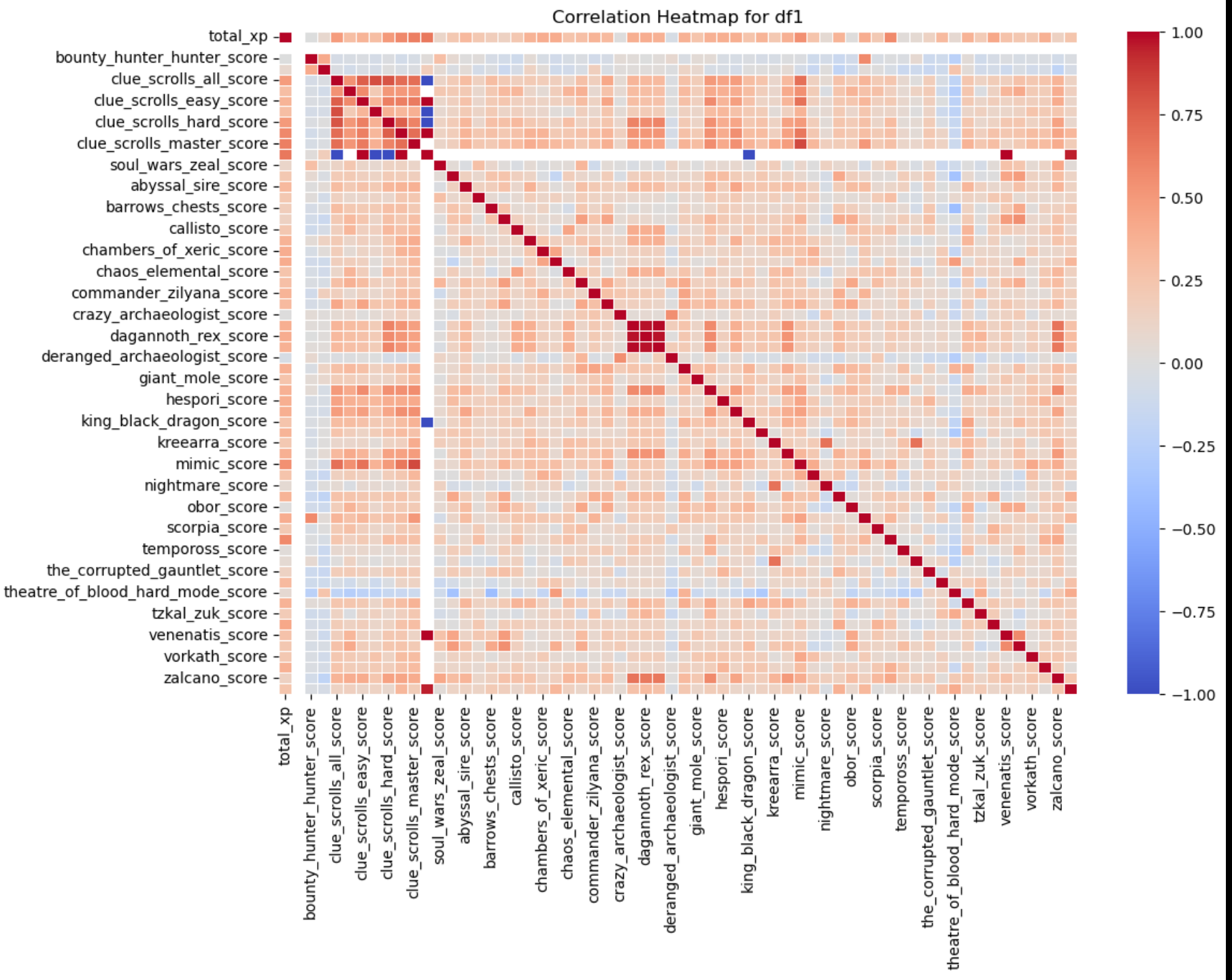
	username	total_xp	unknown_score	bounty_hunter_hunter_score	bounty_hunter_rogue_score	clue_scrolls_all_score	clue_scrolls_beginner_score	clue_scrolls_easy_sco
0	Titus_M	10883717	-1	-1	-1	4	1	
1	Fe Boof	11967413	-1	-1	-1	-1	-1	
2	Vieze Noob	38709020	-1	3	2	-1	-1	
3	0000 1 0000	19987046	-1	7	9	14	1	
4	BIGACLOWN	19727936	-1	-1	-1	3	-1	


Sparsity Heatmap
Rows
0
228
456
684
912
1140
1368
1596
1824
2052
2280
2508
2736
2964
3192
3420
3648
3876
4104
4332
4560
4788
5016
5244
5472
5700
5928
6156
6384
6612
6840
7068
7296
7524
7752
7980
8208
8436
8664
8892
9120
9348
9576
9804
username
total_xp
unknown_score
bounty_hunter_hunter_score
bounty_hunter_rogue_score
clue_scrolls_all_score
clue_scrolls_beginner_score
clue_scrolls_easy_score
clue_scrolls_medium_score
clue_scrolls_hard_score
clue_scrolls_elite_score
clue_scrolls_master_score
pvp_arena_score
soul_wars_zeal_score
rifts_closed_score
abyssal_sire_score
alchemical_hydra_score
barrows_chests_score
bryophyta_score
callisto_score
cerberus_score
chambers_of_xeric_score
chambers_of_xeric_challenge_mode_score
chaos_elemental_score
chaos_fanatic_score
commander_zilyana_score
corporeal_beast_score
crazy_archaeologist_score
dagannoth_prime_score
dagannoth_rex_score
dagannoth_supreme_score
deranged_archaeologist_score
general_graardor_score
giant_mole_score
grotesque_guardians_score
hespori_score
kalphite_queen_score
king_black_dragon_score
kraken_score
kreearra_score
kril_tsutsaroth_score
mimic_score
nex_score
nightmare_score
phosanis_nightmare_score
obor_score
sarachnis_score
scorpia_score
skotizo_score
tempoross_score
the_gauntlet_score
the_corrupted_gauntlet_score
theatre_of_blood_score
theatre_of_blood_hard_mode_score
thermonuclear_smoke_devil_score
tzkal_zuk_score
tztok_jad_score
venenatis_score
vetion_score
vorkath_score
wintertodt_score
zalcano_score
zulrah_score
Columns


Sparsity of Columns (Percentage of Non-Sparse Values)
Percentage of Non-Sparse Values
0
20
40
60
80
100
Columns
unknown_score
pvp_arena_score
theatre_of_blood_hard_mode_score
phosanis_nightmare_score
vetion_score
the_gauntlet_score
chambers_of_xeric_challenge_mode_score
theatre_of_blood_score
deranged_archaeologist_score
nightmare_score
tzkal_zuk_score
nex_score
scorpia_score
chaos_elemental_score
callisto_score
chaos_fanatic_score
venenatis_score
kreearra_score
corporeal_beast_score
sarachnis_score
the_corrupted_gauntlet_score
zalcano_score
crazy_archaeologist_score
kalphite_queen_score
kril_tsutsaroth_score
mimic_score
commander_zilyana_score
soul_wars_zeal_score
alchemical_hydra_score
abyssal_sire_score
grotesque_guardians_score
bryophyta_score
chambers_of_xeric_score
tztok_jad_score
thermonuclear_smoke_devil_score
bounty_hunter_rogue_score
tempoross_score
obor_score
cerberus_score
bounty_hunter_hunter_score
giant_mole_score
dagannoth_prime_score
dagannoth_supreme_score
rifts_closed_score
general_graardor_score
zulrah_score
dagannoth_rex_score
clue_scrolls_master_score
king_black_dragon_score
kraken_score
vorkath_score
skotizo_score
hespori_score
barrows_chests_score
clue_scrolls_elite_score
wintertodt_score
clue_scrolls_beginner_score
clue_scrolls_hard_score
clue_scrolls_all_score
clue_scrolls_medium_score
clue_scrolls_easy_score
total_xp
username


Distribution of Sparsity Across Columns
Number of Columns
0
5
10
15
20
0
20
40
60
80
100
Percentage of Non-Sparse Values

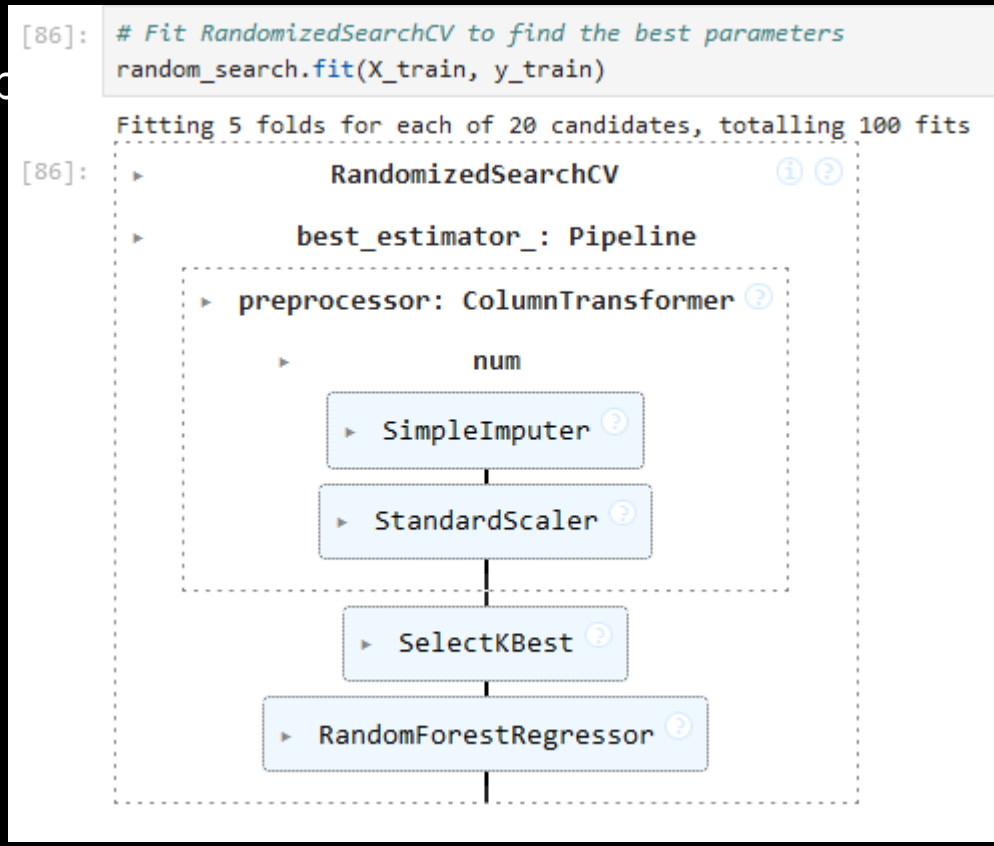

total_xp
unknown_score
bounty_hunter_hunter_score
bounty_hunter_rogue_score
clue_scrolls_all_score
clue_scrolls_beginner_score
clue_scrolls_easy_score
clue_scrolls_medium_score
clue_scrolls_hard_score
clue_scrolls_elite_score
clue_scrolls_master_score
pvp_arena_score
soul_wars_zeal_score
rifts_closed_score
abyssal_sire_score
alchemical_hydra_score
barrows_chests_score
bryophyta_score
callisto_score
cerberus_score
chambers_of_xeric_score
chambers_of_xeric_challenge_mode_score
chaos_elemental_score
chaos_fanatic_score
commander_zilyana_score
corporeal_beast_score
crazy_archaeologist_score
dagannoth_prime_score
dagannoth_rex_score
dagannoth_supreme_score
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grotesque_guardians_score
hespori_score
kalphite_queen_score
king_black_dragon_score
kraken_score
kreearra_score
kril_tsutsaroth_score
mimic_score
nex_score
nightmare_score
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obor_score
sarachnis_score
scorpia_score
skotizo_score
tempoross_score
the_gauntlet_score
the_corrupted_gauntlet_score
theatre_of_blood_score
theatre_of_blood_hard_mode_score
thermonuclear_smoke_devil_score
tzkal_zuk_score
tztok_jad_score
venenatis_score
vetion_score
vorkath_score
wintertodt_score
zalcano_score
zulrah_score


Correlation Heatmap for df1
total_xp
bounty_hunter_hunter_score
clue_scrolls_all_score
clue_scrolls_easy_score
clue_scrolls_hard_score
clue_scrolls_master_score
soul_wars_zeal_score
abyssal_sire_score
barrows_chests_score
callisto_score
chambers_of_xeric_score
chaos_elemental_score
commander_zilyana_score
crazy_archaeologist_score
dagannoth_rex_score
deranged_archaeologist_score
giant_mole_score
hespori_score
king_black_dragon_score
kreearra_score
mimic_score
nightmare_score
obor_score
scorpia_score
tempoross_score
the_corrupted_gauntlet_score
theatre_of_blood_hard_mode_score
tzkal_zuk_score
venenatis_score
vorkath_score
zalcano_score
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

GOAL, HYPOTHESIS, AND METHODS

- Predict 'total_xp' based on 'kill count' from in-game b
- Method will need to be robust to sparse and skewed
- Random Forest Regressor()
- Standard Scaler and Simple Imputation

  strategy = 'constant', fill_value = 0)
- Kbest features
- Random Search instead of Grid Search.



PARAMETER GRID

```
# Define the hyperparameter grid for RandomizedSearchCV
param_dist = {
    'select_k_best__k': [5, 10],  # Try selecting 5 or 10 features
    'rf__n_estimators': randint(50, 200),  # Random number of trees between 50 and 200
    'rf__max_depth': [None, 10, 20],  # Max depth of trees: None, 10, or 20
    'rf__min_samples_split': [2, 5],  # Minimum samples to split a node
    'rf__min_samples_leaf': [1, 2],   # Minimum samples per leaf
    'rf__bootstrap': [True, False],   # Whether to use bootstrap sampling
}
```

BEST MODEL AFTER TRAINING

- Random Forest Hyperparameters:
- rf__bootstrap: True (bootstrap sampling is enabled, meaning random sampling with replacement)
- rf__max_depth: 10
- rf__min_samples_leaf: 2
- rf__min_samples_split: 5 (minimum 5 samples required to split an internal node)
- rf__n_estimators: 160 (the number of trees in the forest is 160)
- select_k_best__k: 10

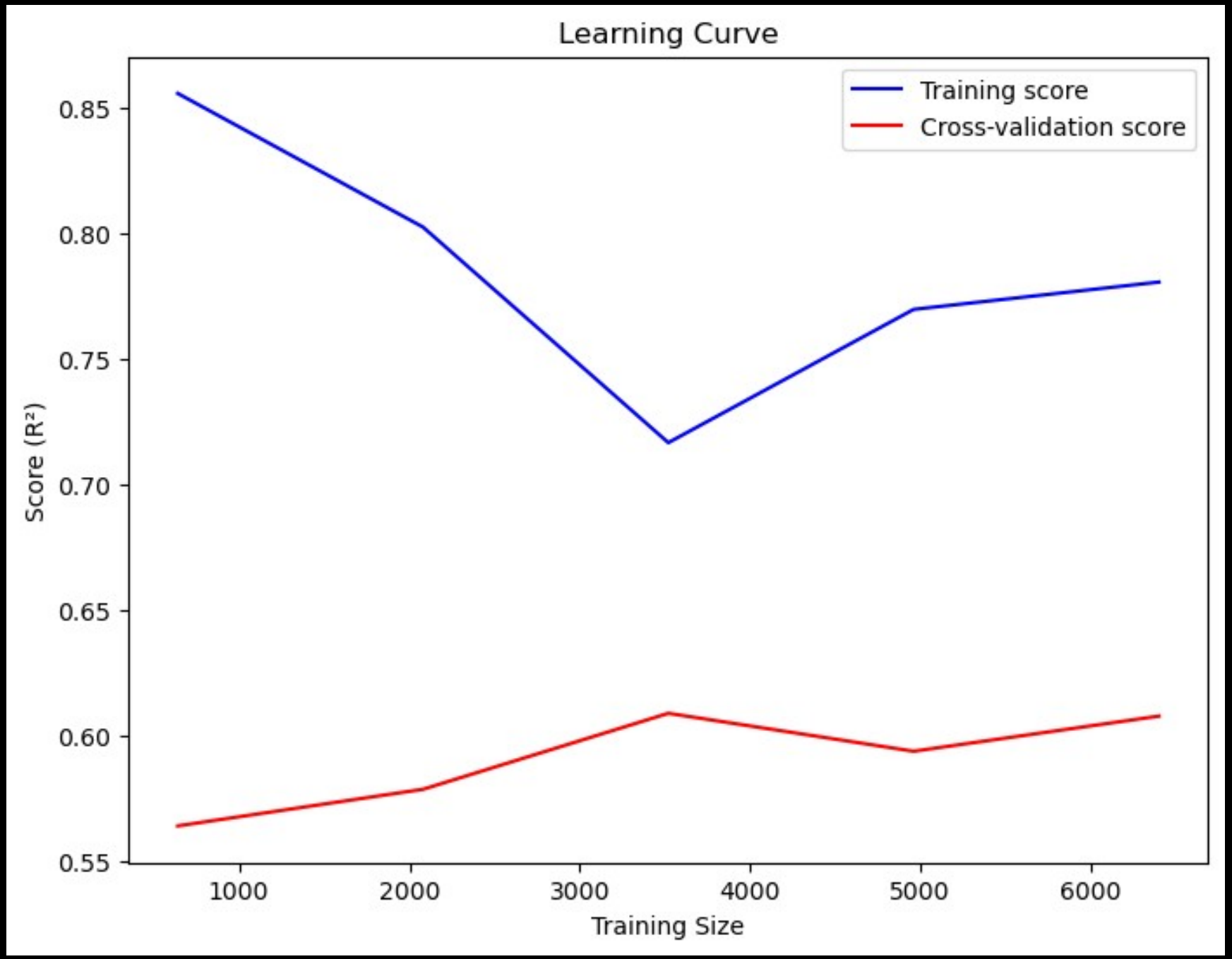
EVALUATION

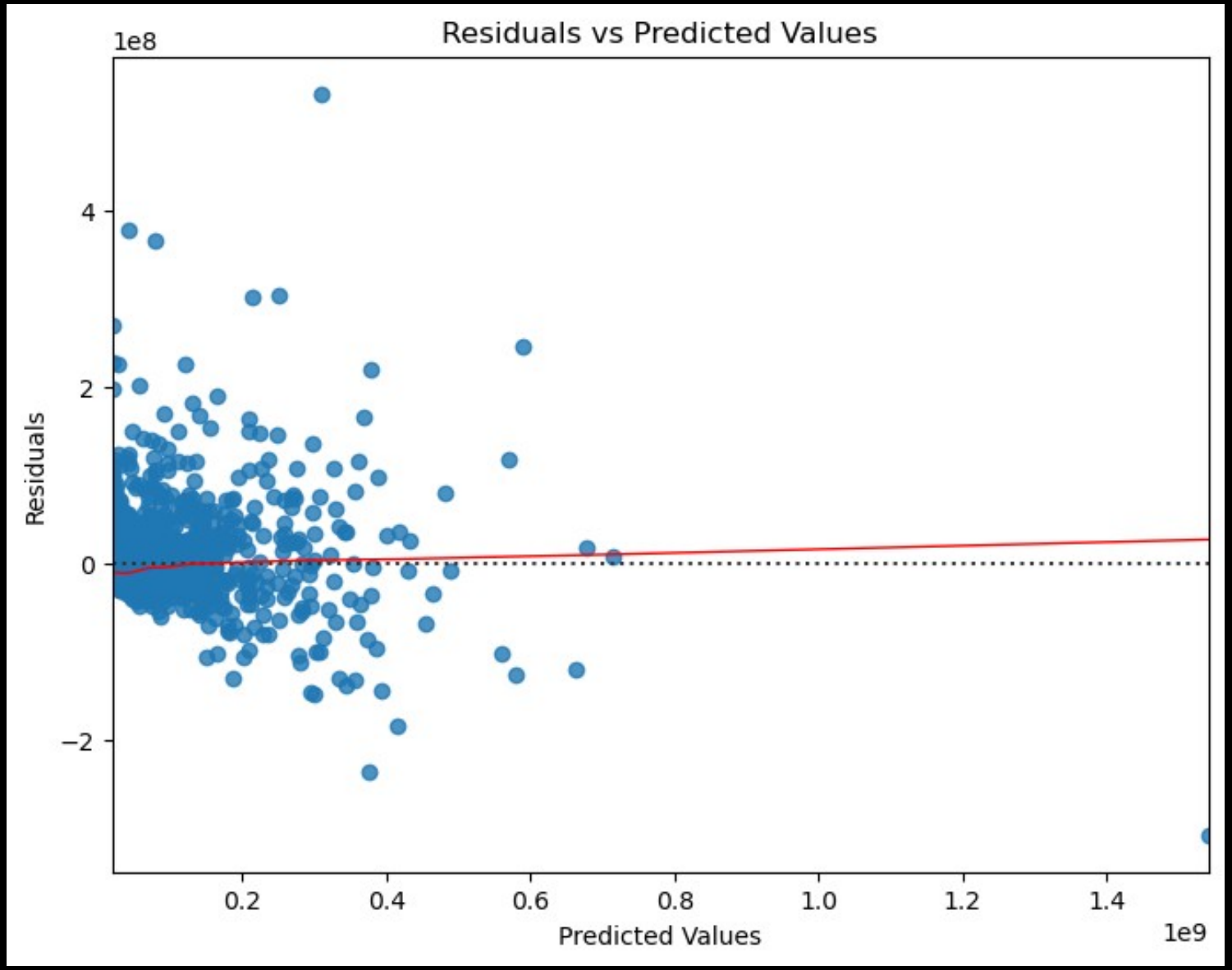
- Cross-Validation Score (Negative MSE): The negative MSE is approximately -6.53e+15. This is a large negative number, but remember that the MSE is the mean squared error, and the smaller the negative value, the better the model fits the data. You can take the absolute value of the negative MSE to interpret it as a regular MSE score, but the key here is the test set R^2 score.

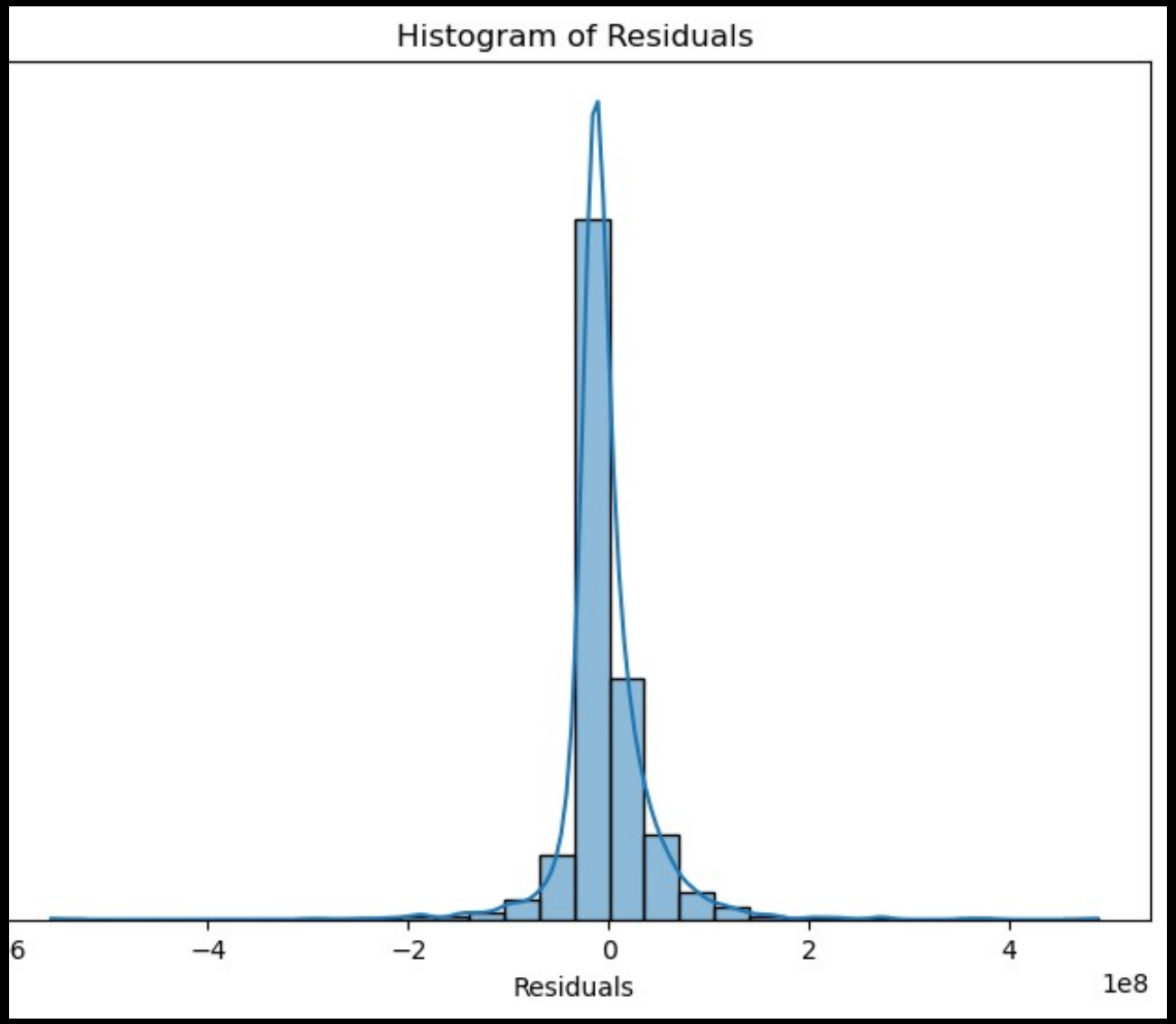
- Test Set Score (R^2): The R^2 score of 0.70 indicates that my model explains about 70% of the variance in the target variable (total_xp). This is a solid performance for a regression task and suggests that the model is doing well, although there's still room for improvement.

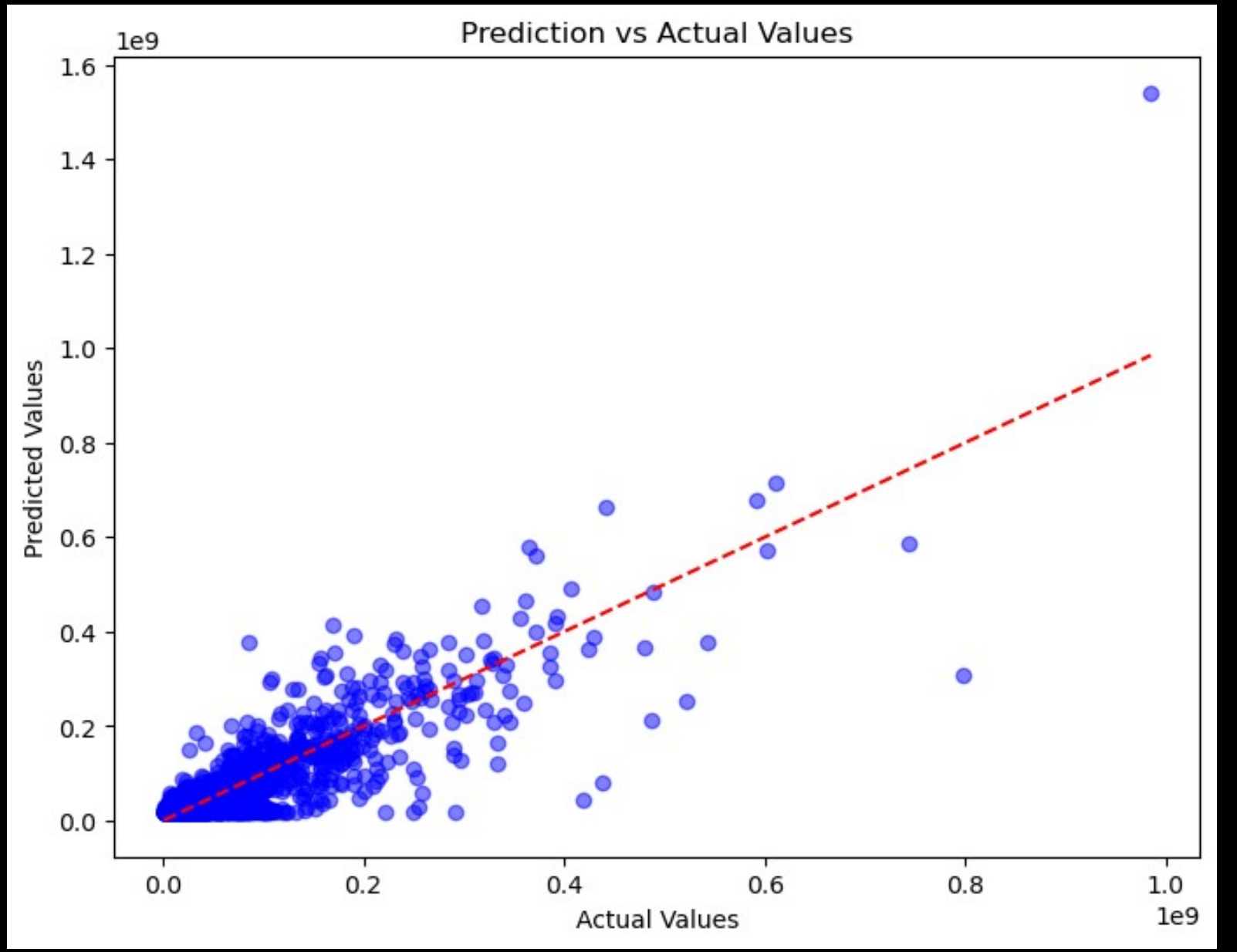
EVALUATION PT. 2

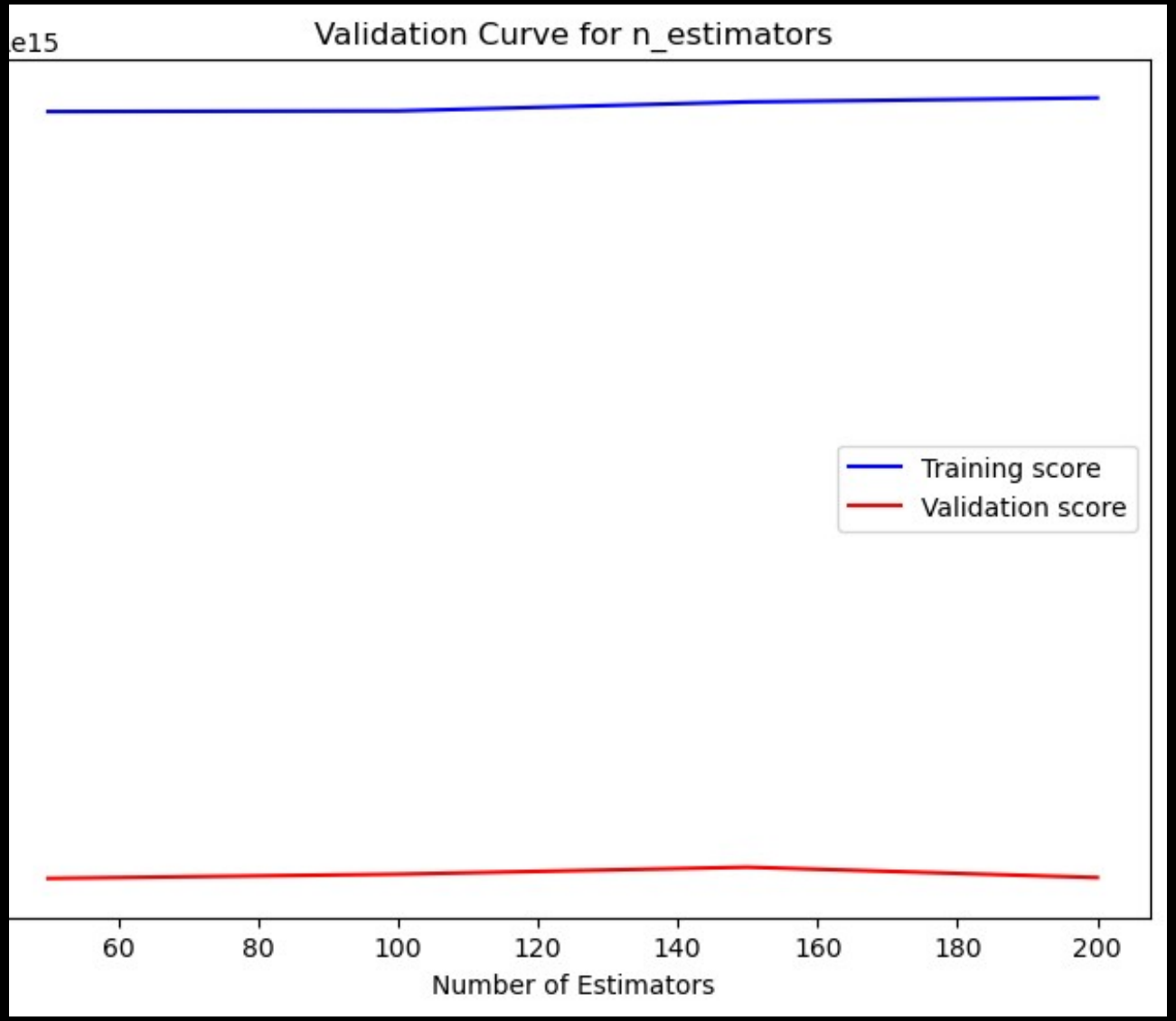
- Negative Mean Squared Error (NMSE): -6537079009911678.0
- Mean Squared Error (MSE): 2095161747615622.8
- Root Mean Squared Error (RMSE): 45772936.847176656
- Mean Absolute Error (MAE): 26740247.982601147
- R^2 Score: 0.6997020818854778


Learning Curve
Training score
Cross-validation score
0.85
0.80
0.75
0.70
0.65
0.60
0.55
Score (R²)
1000
2000
3000
4000
5000
6000
Training Size


Residuals vs Predicted Values
1e8
4
2
0
−2
Residuals
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1e9
Predicted Values


Histogram of Residuals
6
−4
−2
0
2
4
Residuals
1e8


Prediction vs Actual Values
1e9
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Predicted Values
0.0
0.2
0.4
0.6
0.8
1.0
1e9
Actual Values


Validation Curve for n_estimators
e15
Training score
Validation score
60
80
100
120
140
160
180
200
Number of Estimators

C O N C L U S I O N

- Change back to Gridsearch and expand hyperparameters ranges
- Gradient boosting Methods
- Lasso and Ridge Methods
- Grouping-based imputation/ cluster analysis
- Other validation scores(ex. Precision)
- Log transforms
- Feature Extraction and Engineering: Total Boss Kill Count, Low-level Boss Kill Count